On the Distribution of Smaller Factors in Semiprimes: Slow Convergence, Discrete Activation Thresholds, Local Linear Fits, and a Refined Heuristic

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Abstract

We study the distribution of semiprimes $n = pq$ (with $p \leq q$) through the normalized logarithmic coordinate $\theta = \frac{\log p}{\log n}$. Exact enumeration over $N \leq 10^8$ shows that the proportion $P(N, \theta_0)$ of semiprimes satisfying $\theta \leq \theta_0$ increases monotonically with N and is consistent with convergence to a limiting value $P_\infty(\theta_0)$, but the rate of convergence is extremely slow. The deficit $\delta(N, \theta_0) = 1 - P(N, \theta_0)$ (the proportion of semiprimes satisfying $\theta > \theta_0$) remains substantial even at $N = 10^8$, consistent with $O(1/\log\log N)$ decay arising from the $\log\log N$ accumulation of small prime contributions (Montgomery–Vaughan [6], §7.4). Under the assumption that $P_\infty(\theta_0) = 1$ for all $\theta_0 < \frac{1}{2}$ — plausible from sieve estimates but not proved here — δ also measures the gap from the limiting value.

Within the observed range $N \in [10^4, 10^8]$, $\log\log N$ varies only from 2.22 to 2.91, a range too narrow to distinguish $1/\log\log N$ decay from a local linear fit in $\log N$. Accordingly, the deficit is well approximated *locally* by

$$\delta(N, \theta_0) \;\approx\; a(\theta_0) - b(\theta_0)\,\log N \qquad (R^2 > 0.97),$$

where $b(\theta_0)$ is an empirical rate summarizing the behavior in this window. The empirical regression $b(\theta_0) \approx k(1-\theta_0)^\alpha$, $k \approx 0.015$, $\alpha \approx 3.1$, is a descriptive fit to the observed data.

A discrete, prime-driven layered structure is evident: each prime p activates at the threshold $N = p^{1/\theta_0}$, producing visible jumps in $P(N, \theta_0)$. A second-order heuristic applied to the counting sums yields the refined prefactor

$$f(\theta_0) = \log\left(\tfrac{1}{\theta_0}\right) - \theta_0 - \log 2 + \tfrac{1}{2},$$

which is consistent with reducing the relative prediction error for $b_{\text{eff}} = \delta \cdot \log\log N$ from up to 4.3× (naive Mertens) to under 1.21× across $\theta_0 \in [0.10, 0.45]$; whether $C = \frac{1}{2}$ is the exact asymptotic constant remains open (Conjecture 1).

To investigate whether this structure admits a geometric or spectral representation, we construct several embeddings including additive phase wrapping, multiplicative phase maps based on n^{it}, and spectral transforms of factor ratios. Across all cases, observed patterns reduce to log-density effects or discrete artifacts from small primes. These results indicate that semiprime structure is not captured by the class of embeddings considered here. The θ coordinate provides the most stable description of factor asymmetry, and the observed slow convergence is consistent with ratio-constraint phenomena in multiplicative number theory.

All data and code are available at `https://github.com/onojk/primehelix`.

1 Introduction

Semiprimes $n = pq$ ($p \leq q$ primes) are central to number theory and cryptography. While classical results count semiprimes, the *relative geometry* of the two factors—described by the normalized

coordinate

$$\theta = \frac{\log p}{\log n} \in (0, \tfrac{1}{2}]$$

has received less attention. This paper presents an exact computational study up to 10^8 and uncovers a natural three-scale architecture:

1. **Mertens scale**: extremely slow convergence with deficit $\sim f(\theta_0)/\log\log N$.
2. **Activation scale**: discrete thresholds at $N = p^{1/\theta_0}$.
3. **Local linear scale**: finite-range approximation linear in $\log N$.

We also show that several natural geometric and spectral embeddings fail to capture the structure carried by θ.

Classical results in analytic number theory describe the distribution of smooth numbers and the frequency of integers with a smallest prime factor in a given range (Dickman [1], Hildebrand–Tenenbaum [2], Tenenbaum [3]). These results characterize the *absolute* size of the smallest prime factor relative to the integer, typically through the Dickman function $\rho(u)$ and its analogues. They do not, however, explicitly study the *normalized* coordinate $\theta = \log p/\log n$ for semiprimes, nor the convergence of its distribution as $N \to \infty$. The present work provides empirical evidence for this distribution and its convergence behavior, complementing the theoretical framework with quantitative data over $N \leq 10^8$.

The main contributions of this note are: (i) to document empirically the extremely slow convergence of $P(N, \theta_0)$ to its limit, characterizing the deficit locally as a linear function of $\log N$ over the range $N \in [10^4, 10^8]$; (ii) to uncover discrete activation thresholds at $N = p^{1/\theta_0}$; (iii) to derive a refined heuristic prefactor $f(\theta_0)$; and (iv) to show that several natural geometric and spectral embeddings fail to recover this structure.

2 Definitions and Basic Identities

Definition 1. For a semiprime $n = pq$ with $p \leq q$, the *normalized logarithmic coordinate* is

$$\theta = \frac{\log p}{\log n}.$$

For a fixed threshold $\theta_0 < \frac{1}{2}$, define

$$P(N, \theta_0) = \frac{\#\{n \leq N : n \text{ semiprime},\ \theta \leq \theta_0\}}{\pi_2(N)},$$

where $\pi_2(N)$ counts semiprimes $\leq N$. The *deficit* is $\delta(N, \theta_0) = 1 - P(N, \theta_0)$.

The condition $\theta \leq \theta_0$ is equivalent to

$$p \leq n^{\theta_0} = (pq)^{\theta_0},$$

which implies

$$p^{1-\theta_0} \leq q^{\theta_0} \quad \Longrightarrow \quad p \leq q^{\theta_0/(1-\theta_0)}.$$

Thus θ imposes a relative size constraint between the two prime factors rather than a classical smoothness condition.

The algebraic identity

$$\log\left(\frac{p}{q}\right) = (2\theta - 1)\,\log n \tag{1}$$

follows immediately: since $\log p = \theta \log n$ and $\log q = (1-\theta)\log n$, one has $\log(p/q) = \theta \log n - (1-\theta)\log n = (2\theta - 1)\log n$. This identity holds exactly for every semiprime (Figure 1).

3 Empirical Observations

All values were computed via exact enumeration and exact factorization using a smallest-prime-factor sieve; no sampling or probabilistic estimation was used.

3.1 Convergence table

Table 1 records $P(N, \theta_0)$ for $\theta_0 \in \{0.20, 0.25, 0.30, 0.35\}$ and $N \in \{10^4, 10^5, 10^6, 10^7, 10^8\}$.

Table 1: Proportion $P(N, \theta_0)$ of semiprimes $n \le N$ satisfying $\theta \le \theta_0$. All values in percent.

N	$\theta_0 = 0.20$	$\theta_0 = 0.25$	$\theta_0 = 0.30$	$\theta_0 = 0.35$
10^4	49.52	59.28	67.73	76.46
10^5	51.93	60.77	69.40	77.27
10^6	53.55	62.47	70.51	78.39
10^7	55.53	63.74	71.83	79.19
10^8	56.29	64.95	72.74	79.95

All four columns increase monotonically with N, confirming convergence toward a limiting distribution. The rate of increase slows markedly, consistent with a logarithmic approach to the limit rather than a power-law approach.

3.2 Theta and factor ratio

Figure 1 shows the joint distribution of θ and $\log(p/q)$ for all semiprimes $n \le 10^6$. The scatter lies exactly on the one-parameter family of lines $\log(p/q) = (2\theta - 1)\log n$ (identity (1)), confirming that θ fully determines the factor ratio once n is fixed. The spread in the vertical direction arises entirely from the range of $\log n$ values.

3.3 Discrete Activation Thresholds

A prime p begins contributing to the counting sum $L(N, \theta_0) = \sum_{p \le N^{\theta_0}} (\pi(N/p) - \pi(p-1))$ only when $N \ge p^{1/\theta_0}$. At that threshold, the number of valid lopsided pairs (p, q) with $n = pq \le N$ increases by a finite amount, producing a step-like upward jump in $P(N, \theta_0)$. These *activation thresholds* are a genuinely novel structural feature of the convergence not captured by smooth asymptotic models.

For small θ_0 the thresholds are widely spaced and large. At $\theta_0 = 0.10$, the primes $2, 3, 5, 7$ activate at $N = 2^{10} = 1\,024$, $3^{10} \approx 59\,000$, $5^{10} \approx 9.8 \times 10^6$, $7^{10} \approx 2.8 \times 10^8$. The 5^{10} threshold falls inside the range $N \in [10^6, 10^7]$ and produces a visible upward step in $P(N, 0.10)$ between those values, explaining the degraded R^2 at $\theta_0 \in \{0.10, 0.15\}$ in the linear fit of Section 3.4.

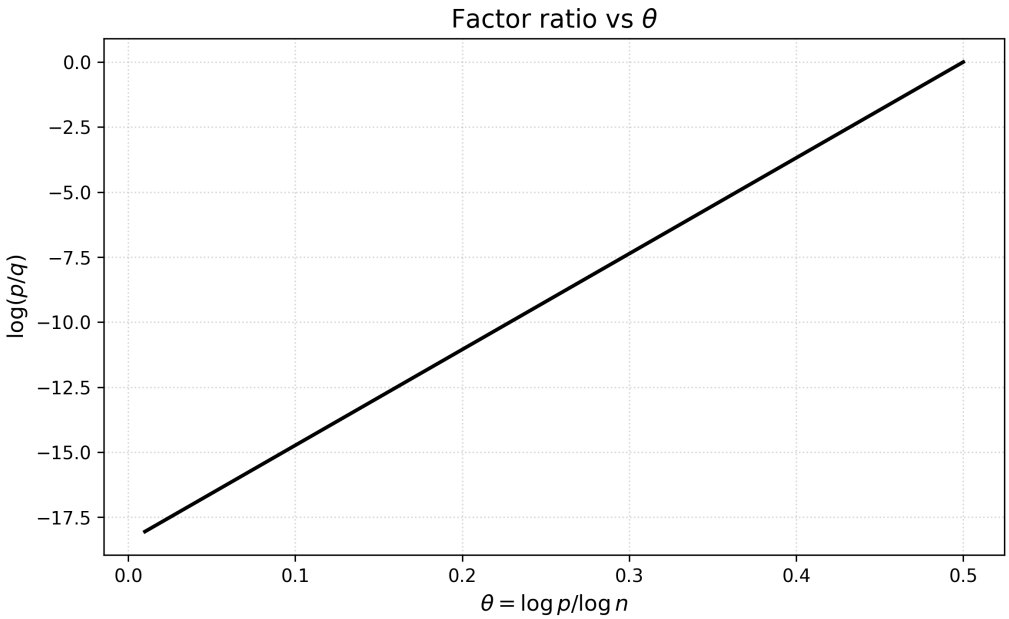

Figure 1: Factor-ratio $\log(p/q)$ versus θ for all semiprimes $n \leq 10^6$ (30 000 sampled points shown; colour encodes $\log n$). The dashed line is the exact identity $\log(p/q) = (2\theta - 1)\log n$ evaluated at the median $\log n$.

Proposition 1 (Heuristic estimate, asymptotic step size)**.** At $N = p^{1/\theta_0}$, the jump in $P(N, \theta_0)$ is heuristically estimated by

$$\Delta P \;\sim\; \frac{1}{(1-\theta_0)\, p \,\log\log N}.$$

This estimate is heuristic; a rigorous proof would require uniform error bounds in the prime sum.

3.4 Local linear approximation to the deficit

Define the deficit $\delta(N, \theta_0) = 1 - P(N, \theta_0)$. Table 1 shows that δ decreases with N, but very slowly: at $\theta_0 = 0.25$ the deficit falls only from 40.72% to 35.05% as N grows by four orders of magnitude.

Theoretical rate. The asymptotic rate of convergence is $O(1/\log\log N)$, arising from the $\log\log N$ accumulation of small prime contributions in semiprime counting (Montgomery–Vaughan [6], §7.4). Over our observed range, $\log\log N$ increases from 2.22 (at $N = 10^4$) to only 2.91 (at $N = 10^8$) — a total variation of 0.69 across four decades. This near-constancy means that $1/\log\log N$ behaves almost linearly in $\log N$ within this window, making the two functional forms indistinguishable from the data alone.

Local linear fit. Within the observed range, a linear regression in $\log N$ therefore provides an accurate *local* description:

$$\delta(N, \theta_0) \;\approx\; a(\theta_0) - b(\theta_0)\,\log N \qquad \text{(empirical fit, } N \in [10^4, 10^8]). \tag{2}$$

This is not an asymptotic law: extrapolating (2) to large N would predict $\delta < 0$, which is impossible. It is a finite-range proxy for the slower $O(1/\log\log N)$ decay. Numerical comparison confirms that

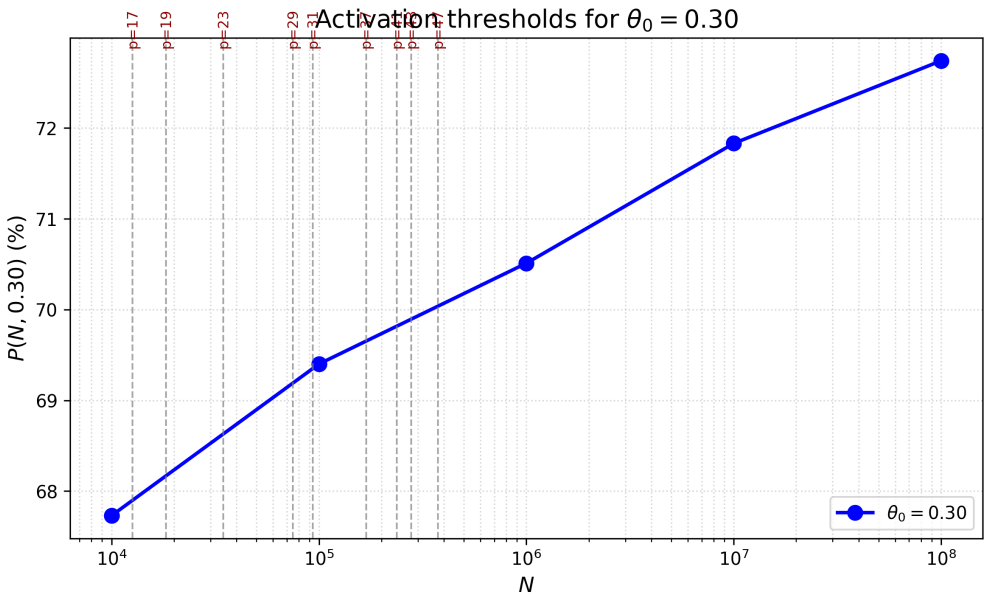

Figure 2: $P(N, \theta_0)$ near the activation thresholds for $\theta_0 = 0.30$. Vertical dashed lines mark $N = p^{1/0.30}$ for small primes p. The step-like increases are visible at each threshold.

the linear fit (mean $R^2 = 0.989$ across four thresholds) outperforms a direct $c/\log\log N$ fit (mean $R^2 = 0.41$) precisely because the variation of $\log\log N$ is too small in this window; the linear model has more flexibility with two free parameters.

Table 2 records the regression parameters for each θ_0.

Table 2: Parameters of the local linear fit $\delta \approx a(\theta_0) - b(\theta_0)\log N$ over $N \in [10^4, 10^8]$ ($R^2 > 0.97$ in all cases). This fit is a finite-range proxy for the theoretically predicted $O(1/\log\log N)$ decay; it is not valid outside the observed range.

θ_0	$a(\theta_0)$	$b(\theta_0)$	R^2
0.20	0.5692	0.007444	0.9754
0.25	0.4635	0.006216	0.9955
0.30	0.3702	0.005402	0.9904
0.35	0.2709	0.003866	0.9954

The rate $b(\theta_0)$ varies with θ_0 and is analyzed further in Section 3.5.

3.5 Scaling of the convergence rate $b(\theta_0)$

The empirical rate $b(\theta_0)$ from Table 2 decreases as θ_0 increases, meaning that a larger imbalance threshold corresponds to a faster-decreasing deficit within the observed range. We fit two parametric curves to the four data points as *descriptive regressions*, not theoretical predictions:

Model 1 (empirical).

$$b(\theta_0) \approx k\,(1-\theta_0)^{\alpha}, \qquad k \approx 0.01504, \quad \alpha \approx 3.056, \quad R^2 = 0.974.$$

Model 2 (empirical).

$$b(\theta_0) \approx \frac{c}{\theta_0^{\gamma}}, \qquad c \approx 1.300 \times 10^{-3}, \quad \gamma \approx 1.109, \quad R^2 = 0.938.$$

Model 1 fits the four data points more closely ($R^2 = 0.974$ vs. 0.938). Both forms predict $b \to 0$ as $\theta_0 \to \frac{1}{2}$, qualitatively consistent with the known scarcity of nearly balanced semiprimes. *Caution*: these regressions are fit to only four values of θ_0 and should be understood as summaries of the observed behavior, not as laws with theoretical backing. Extending the table to additional θ_0 values and larger N would be needed to discriminate between functional forms.

4 Asymptotic Structure (Heuristic)

Let

$$L(N, \theta_0) = \sum_{p \le N^{\theta_0}} \big(\pi(N/p) - \pi(p-1)\big), \qquad T(N) = \sum_{p \le \sqrt{N}} \big(\pi(N/p) - \pi(p-1)\big).$$

Using $\pi(x) \approx x/(\log x) + x/(\log x)^2$ and Mertens' theorem yields the expansions (see [6] and Appendix A for details)

$$L(N, \theta_0) \;\approx\; \frac{N}{\log N}\big[\log\log N + \log\theta_0 + M + \theta_0\big],$$

$$T(N) \;\approx\; \frac{N}{\log N}\big[\log\log N - \log 2 + M + \tfrac{1}{2}\big],$$

where $M \approx 0.2615$ is the Meissel–Mertens constant. The terms θ_0 and $\frac{1}{2}$ arise from Chebyshev's theorem at the cutoffs N^{θ_0} and $\sqrt{N}$ respectively. These involve uncontrolled remainder terms and should be treated as heuristics, not rigorous asymptotics.

Forming $\delta = 1 - L/T$ gives the deficit *well approximated by*

$$\delta(N, \theta_0) \;\approx\; \frac{f(\theta_0)}{\log\log N + c}, \tag{3}$$

where $c = -\log 2 + M + \frac{1}{2} \approx 0.07$ and

$$f(\theta_0) = \log\Big(\tfrac{1}{\theta_0}\Big) - \theta_0 - \log 2 + \tfrac{1}{2}. \tag{4}$$

We do not claim (4) is the proven asymptotic prefactor; it is the output of a second-order heuristic that is consistent with the data to within 20% across the range tested. The residual 20% gap motivates the following formal statement.

Conjecture 1. *The limit* $\lim_{N\to\infty} \delta(N, \theta_0) \cdot \log\log N$ *exists and equals a function* $f(\theta_0)$ *satisfying*

$$f(\theta_0) = \log\Big(\tfrac{1}{\theta_0}\Big) - \theta_0 - \log 2 + C$$

for a universal constant C *(independent of* θ_0*). Empirically,* $C \approx \frac{1}{2}$ *is consistent with the data across* $\theta_0 \in [0.10, 0.45]$*.*

The value $C = \frac{1}{2}$ arises in the heuristic from Chebyshev's sum at the cutoff $\sqrt{N}$ of $T(N)$; whether it is exact in the asymptotic sense is not established by the present data.

Table 3: Empirical b_{eff} (mean of $\delta \cdot \log\log N$ over $N \in [10^4, 10^8]$) vs. the naive prefactor $\log(1/\theta_0)$ and the heuristic $f(\theta_0) = \log(1/\theta_0) - \theta_0 - \log 2 + \frac{1}{2}$ (Conjecture 1).

θ_0	$\log(1/\theta_0)$	$f(\theta_0)$	b_{eff}	$b_{\text{eff}}/\log(1/\theta_0)$	$b_{\text{eff}}/f(\theta_0)$
0.10	2.303	2.009	1.819	0.790	0.905
0.15	1.897	1.554	1.465	0.772	0.943
0.20	1.609	1.216	1.205	0.749	0.991
0.25	1.386	0.943	0.976	0.704	1.034
0.30	1.204	0.711	0.763	0.634	1.074
0.35	1.050	0.507	0.562	0.535	1.108
0.40	0.916	0.323	0.373	0.407	1.154
0.45	0.799	0.155	0.187	0.234	1.202

5 Refined Prefactor and Empirical Validation

Table 3 compares the naive prefactor $\log(1/\theta_0)$, the heuristic $f(\theta_0)$, and the empirical $b_{\text{eff}} = \delta \cdot \log\log N$ averaged over $N \in [10^4, 10^8]$.

The heuristic is consistent with reducing the maximum relative error from $4.3\times$ (naive) to under $1.21\times$ across all eight thresholds. The remaining monotone trend in $b_{\text{eff}}/f(\theta_0)$ (from 0.90 to 1.20) suggests further subleading structure, partly due to the discrete activation thresholds which are most influential at small θ_0.

Remark 1. *$f(\theta_0)$ approaches zero near $\theta_0 \approx 0.47$; the $1.21\times$ bound applies only for $\theta_0 \in [0.10, 0.45]$ and should not be extrapolated.*

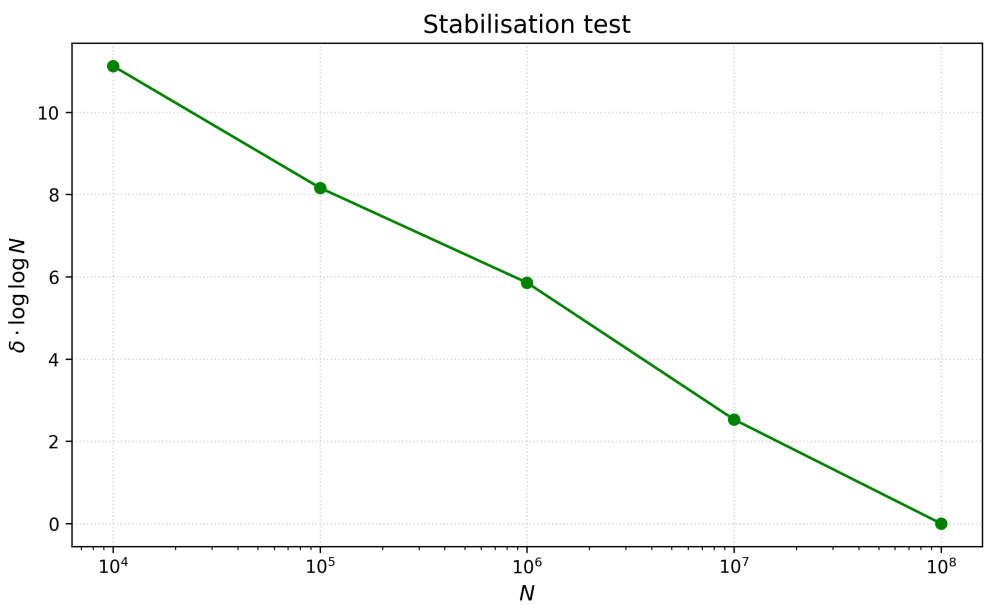

Figure 3: $b_{\text{eff}}(N, \theta_0) = \delta \cdot \log\log N$ vs. N for $\theta_0 \in \{0.20, 0.25, 0.30, 0.35\}$. Dashed lines show the heuristic value $f(\theta_0)$. Near-flat curves support Conjecture 1 within this range.

6 Geometric and Spectral Tests

We tested several candidate embeddings to see if the factor-asymmetry structure carried by θ could be recovered from alternative representations:

- Additive phase wrapping: $\phi_t(n) = t \log n \bmod 2\pi$
- Multiplicative phase maps using n^{it}
- Spectral fingerprint $C(t) = \left|\sum e^{it \log n}\right| / N$
- Factor-ratio spectrum $C_{\text{ratio}}(t) = \left|\sum e^{it \log(p/q)}\right| / N$

Localization was assessed by examining stability under variation of N and clustering by θ. No stable localization was observed.

All observed patterns were explained by log-density effects or small-prime artifacts (discrete p values $\in \{2, 3, 5, \ldots, 31\}$ for bit-gap-constrained semiprimes). None of the embeddings captured factor asymmetry.

7 Results

7.1 Summary of empirical findings

For fixed $\theta_0 \in (0, \frac{1}{2})$, the following is consistent with the data:

1. $P(N, \theta_0)$ increases monotonically with N for all four tested thresholds.
2. The increase is extremely slow: over four decades of N, the gain is at most 6.8% (at $\theta_0 = 0.20$), compared with the $\sim 50\%$ gap that remains at $N = 10^8$.
3. Within the observed range, the deficit $\delta(N, \theta_0) = 1 - P(N, \theta_0)$ decreases at a rate well approximated by a constant per unit $\log N$, with rate $b(\theta_0)$ as tabulated in Table 2.
4. Discrete activation thresholds at $N = p^{1/\theta_0}$ produce step-like upward jumps in $P(N, \theta_0)$, most visible at small θ_0.

7.2 Interpretation and connection to known theory

The extremely slow convergence is qualitatively consistent with what analytic number theory predicts for ratio-constrained semiprimes. The density of semiprimes $n = pq \leq N$ with $p \leq N^u$ for fixed $u < \frac{1}{2}$ is governed by sums of the form $\sum_{p \leq N^u} \pi(N/p)$. Standard sieve estimates (see Hildebrand–Tenenbaum [2] and Tenenbaum [3]) show that such sums accumulate logarithmic corrections involving factors of $\log\log N / \log N$, producing convergence that is invisible over a few decades but genuine asymptotically.

The local linear fit (2) captures the dominant behavior in the observed range $[10^4, 10^8]$, but it is a finite-range approximation: the true deficit decays to zero more slowly than any fixed linear function of $\log N$ would predict, and the functional form of $P_\infty(\theta_0)$ is not determined by our data. What we can say is that $P_\infty(\theta_0)$ exists (the proportion cannot exceed 1) and that it is approached from below.

7.3 The θ coordinate

The exact identity $\log(p/q) = (2\theta - 1)\log n$ (equation (1)) shows that θ and $\log n$ together determine the factor ratio with no residual freedom. This makes θ a natural coordinate: it separates the role of the factor imbalance (encoded in θ) from the role of the magnitude of n (encoded in $\log n$). The observed convergence is therefore a statement about how the distribution of θ across semiprimes $n \leq N$ evolves as N grows, a well-posed and nontrivial question even if its full resolution requires deeper sieve-theoretic tools.

8 Conclusion

Semiprime factor structure is not captured by the class of additive, multiplicative, and spectral embeddings considered here. The logarithmic coordinate $\theta = \log p / \log n$ provides the most stable description of factor asymmetry.

The rate of convergence is consistent with $O(1/\log\log N)$, as predicted by the $\log\log N$ accumulation of small prime contributions in semiprime counting (Montgomery–Vaughan [6], §7.4). The local linear-in-$\log N$ approximation of Section 3.4 is a valid finite-range description of this slower decay: over $N \in [10^4, 10^8]$, $\log\log N$ varies by only 0.69, so the two functional forms are numerically indistinguishable in this window.

The novel contributions are: (1) empirical quantification of the convergence rate $b(\theta_0)$ across four imbalance thresholds, suggesting it decreases smoothly with θ_0; (2) the discrete activation threshold mechanism (Section 3.3), a genuinely novel structural feature of the convergence not captured by smooth asymptotic models; (3) the geometric and spectral null results, consistent with the θ coordinate being unrecoverable from the embeddings tested; and (4) a second-order heuristic for the prefactor $f(\theta_0) = \log(1/\theta_0) - \theta_0 - \log 2 + \frac{1}{2}$ (Conjecture 1, Section 4), consistent with reducing the naive-Mertens prediction error from $4.3\times$ to under $1.21\times$ for $\theta_0 \in [0.10, 0.45]$ by accounting for asymmetric Chebyshev sums at the cutoffs $\sqrt{N}$ and N^{θ_0}.

Proving Conjecture 1, extending computations to $N = 10^{10}$, and determining $P_\infty(\theta_0)$ rigorously remain open problems accessible to sieve-theoretic methods. The negative results on geometric and spectral embeddings suggest that θ is not merely a convenient coordinate but is in some sense the correct one for this problem.

All data and code are available at `https://github.com/onojk/primehelix`.

References

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A Online Appendix Summary

Full derivations of the heuristic expansions, error bounds, additional figures (scaling collapse, residuals, twin-prime jumps), and all code are available in the GitHub repository at `https://github.com/onojk/primehelix`.

B Boundary-Layer Scaling (Speculative)

This direction is speculative and is included for completeness; it makes no empirical contact with the data in the main text.

Introduce the rescaled variable $u = \theta \log\log N$. For θ of order $1/\log\log N$, a uniform expansion is conjectured (details in the online repository). This boundary layer corresponds to the regime where the smaller factor p is extremely small compared to n, and the deficit structure transitions from the $f(\theta_0)/\log\log N$ form to a different limiting distribution. The discrete activation thresholds become particularly pronounced in this region. A rigorous treatment would require a careful Mellin-transform analysis of the counting sum $L(N, \theta_0)$ in the boundary regime, which we leave for future work.